Circuit electricity and the quark model as gauge theories

Is the quark model circuit electricity at the quantum level? — register entry H22 (the user's "quantumtricity" hypothesis)

kronspice simulation report · 2026-11-07 · work in progress

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Summary

Register entry H22 holds the user's hypothesis that the Ohm's-law wheel (P, V, I, R) and the Standard Model wheel describe one structure, the quark model being the principles of electricity at the quantum level. As stated it makes no prediction, so it is tested in a definite form. The one established link is that both are gauge theories on a graph: circuit theory is the U(1) (abelian) theory on Kron's network, and the lattice quark model is the SU(3) (non-abelian) theory on the same kind of graph. Step 1 checks the abelian dictionary; step 2 searches for any circuit quantity that composes the non-abelian way.

- **Step 1 passes (standard):** on a two-mesh RLC circuit in Kron's mesh frame, KVL is zero holonomy of the branch voltage (residual 2×10^{-16}) and KCL is the rate of the Gauss generator (10^{-17}).
- **The U(1) lattice field is Yee's FDTD:** on a 21×17×6 grid it equals *em.FDTD* edge by edge to 6.6×10^{-13} of max E over 4000 steps of the TM110 mode; energies agree to 8.6×10^{-15} of W_0.
- **Step 2 is null so far:** SU(3) mesh loops compose with defect 0.75 on average (min 0.37 over 200 random fields), 0 for the abelian (Cartan) part; every circuit quantity composes with defect ≤ 3×10^{-16}, the round-off of addition, in the linear circuit and in a diode pair.
- **The quantum level of V, I and R is electrons and photons:** $K_J = 2e/h$ = 483 597.848 4 GHz/V, $R_K = h/e^2$ = 25 812.807 459 Ω, I = e·f — no quark content.
- **Status: bounded, work in progress.** The matter sector (quark ↔ electron at the nodes) is not yet modelled.

What is being tested

The hypothesis (H22)

The quark model and the principles of circuit electricity (V, I, R, P) are one structure, the latter being electricity at the quantum level (user hypothesis, 2026-11-01). The name "quantumtricity" is the user's and is used for this example only, never as an established term.

The testable form and the standard prediction

- **Prediction of the hypothesis:** a circuit quantity exists that composes as SU(3) link variables do, i.e. one whose loop products around two meshes meeting at a node do not commute.
- **Standard prediction:** none exists. Circuit theory is the classical, abelian (U(1)) limit of a lattice gauge theory and the quark model is the non-abelian (SU(3)) one on the same kind of graph; every Kirchhoff sum commutes, so the resemblance ends at "both are gauge theories on a network". The quantum level of V, I and R is set by electrons and photons through e and h; conduction current is electrons only, and quarks fix only the lattice charge Z.

The dictionary: both are gauge theories on a graph

A lattice gauge theory (Wilson 1974) has a group element on every link of a graph, a gauge transformation at every node, and gauge-invariant quantities on closed loops. Kron's network has the same three objects: the branch, the junction and the mesh. *kron/gauge.go* writes the correspondence out (A is Kron's incidence tensor, C his connection tensor):

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Lattice gauge theory	Kron's network (kron/)
node, link, plaquette	junction, branch, mesh
Grad χ (node → link)	E = A·v, the incidence tensor
Curl a (link → plaquette), holonomy	$e' = C^T \cdot e$, the loop sum of the connection tensor
Curl(Grad) = 0	$A^T \cdot C = 0$ (connection.go)
Div = Grad^T	$A^T \cdot i$, Kirchhoff's current law
Curl^T	$i = C \cdot i'$, branch currents from mesh currents
Gauss generator Div Q − ρ	node charge balance; its time derivative is KCL
pure gauge a = Grad χ, zero holonomy	the branch voltage, E = Grad(v_{node}): KVL
Hamiltonian $\Sigma\, Q^2/2C + \Sigma\, \Phi_p{}^2/2L_p$	the ledger's ½Cv² and ½Li² (mesh current $i_p = \Phi_p/L_p$)
Wilson loop $\exp(iq\Phi_p/\hbar)$	mesh flux linkage in flux quanta

For the group U(1) this is Maxwell's electrodynamics on Kron's 1944 field network (Tonti 2001 is the modern statement), and in the lumped limit it is circuit theory. For SU(3) it is the lattice form of the quark model (Kogut & Susskind 1975).

Step 1 — abelian (standard): circuit theory is U(1) on Kron's network

A two-mesh RLC circuit (V1 R1 L1 L2 R2 C1, three fundamental meshes) is solved in Kron's mesh frame at 1 kHz, and the identities are read off the solution:

Identity	Residual
Curl(Grad) = 0 ($A^T \cdot C = 0$)	0
KVL as zero holonomy of the branch voltage	2×10^{-16}
the voltage is pure gauge, E = Grad(v_{node})	2×10^{-16}
mesh EMF = −jω × flux holonomy (Faraday with the flux in the inductors)	1×10^{-16}
KCL as the rate of the Gauss generator, Div I = 0	1×10^{-17}
continuity, Div I_C = jω·Div Q = jω·ρ_{node}	1×10^{-18}

Mesh 0 links 6.1×10^{-5} Wb at the phasor peak, 2.9×10^{10} flux quanta: a Cooper pair carried round it would have its Wilson loop turned through 2π that many times. The Hamiltonian's two terms are the ledger's energies element by element (C1: ½C|v|², L1, L2: ½L|i|²).

The same field on Kron's 1944 field network against FDTD

The same U(1) field on a 21×17×6 Yee grid closed by conducting walls is stepped beside *em.FDTD* on the TM110 cavity mode for 4000 steps:

Lattice vs FDTD	Agreement
E, edge by edge	6.6×10^{-13} of max E
energy: lattice Hamiltonian vs FDTD ledger	8.6×10^{-15} of W_0
exact discrete energy drift	2.0×10^{-14}
Gauss generator at interior nodes	0
ω against Yee's own dispersion / the continuum	1.1×10^{-7} / 5.2×10^{-4}

They agree to round-off because Hamilton's equations on the lattice, $dQ/dt = \text{Curl}^T(\Phi/L)$ and $da/dt = -Q/C$, are Yee's leapfrog. The field network and the circuit are one construction: a branch capacitor is an edge's ε·(dual face)/length, a mesh inductance is a face's μ·(face)/(dual length).

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Step 2 — non-abelian (the hypothesis): SU(3) on the same graph

kron.SU3Field puts an SU(3) element on every branch of the same two-mesh graph. Gauge transformations $U_k \to g_{from}\, U_k\, g_{to}^\dagger$ leave the Wilson-loop traces unchanged (6×10^{-15}) and turn the Gauss generator covariantly ($G_n \to g_n\, G_n\, g_n^\dagger$, 10^{-14}) but not invariantly: it changes by order 1, because it is a matrix, not a number. Two meshes meeting at a node compose by matrix multiplication; the composition defect is $\|W_pW_q - W_qW_p\|/3$ for loop elements W_p, W_q:

Field on the links	Composition defect
random SU(3), 200 fields	mean 0.75, min 0.37
diagonal (Cartan) SU(3), the abelian U(1)×U(1) part	0

A circuit quantity that composed as SU(3) currents do would show such a defect. The search covers every quantity the simulator carries on a branch:

Circuit	Quantities	Largest composition defect
two-mesh RLC, Kron mesh frame	voltage, current, charge, flux, complex power	3×10^{-16} (the round-off of addition)
diode pair, operating point (D1 0.669 V 1.73 mA, D2 0.652 V 0.87 mA)	voltage, current; KCL 9×10^{-15}, KVL 0	0

None found. Every Kirchhoff sum commutes; a nonlinear constitutive law (the diodes) changes the numbers, not the composition law. The difference between the two wheels is exactly the difference between U(1) and SU(3): the circuit is the abelian, classical limit.

The quantum level of V, I and R

Since the 2019 SI the three units of the Ohm's-law wheel are realized from the exact e and h (computed in the run from CODATA 2018):

Unit	Realization	Source
volt	$K_J = 2e/h$ = 483 597.848 4 GHz/V	Josephson 1962
ohm	$R_K = h/e^2$ = 25 812.807 459 Ω	von Klitzing, Dorda & Pepper 1980
ampere	$I = e{\cdot}f$; 1 nA = 6.24×10^9 electrons/s	Pekola et al. 2013

- One lepton and one gauge boson of the Standard Model wheel; no quark.
- The only quark content a circuit sees is the lattice charge Z of its conductors (copper: 29 × uud = 29 e). Conduction current is electrons.
- The fractional charges e/3 and e/5 measured in the fractional quantum Hall effect (de-Picciotto et al. 1997; Saminadayar et al. 1997) are collective excitations of electrons, not quarks.

Limits and open work

- **The matter sector.** The dictionary matches the gauge fields only (photon ↔ gluon). A quark is a charged matter field at the nodes; its circuit counterpart is the electron, entering the Gauss law as the node charge ρ. Step 2 carried the quark side only as abstract colour charges ρ_n. A charge field on the nodes with its hopping current along the links is the next construction.
- **Where the wheel's terms land.** V is the electric field on a link, I the matter current along it, P = V·I the work term J·E between matter and field, and R has no counterpart: a gauge theory is lossless, and resistance is coupling to a heat bath. Still to be written into the dictionary with a test.
- **The TLM twin.** The register asks for agreement with *em/tlm.go* as well as *em/fdtd.go*; TLM is not yet written.

Status

Register entry H22: **bounded**, work in progress. Step 1 passes; step 2 is a null result so far, which records a bound. The entry stays in the register and the example keeps growing. What would move it: a circuit quantity with a non-zero composition defect, i.e. one that carries an index transported along branches and whose node sums do not commute. None of the simulator's quantities does.

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Reproducing the results

```
go run ./examples/quantumtricity
go test ./examples/quantumtricity
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Sources: K. G. Wilson, *Phys. Rev. D* 10, 2445 (1974); J. Kogut, L. Susskind, *Phys. Rev. D* 11, 395 (1975); G. Kron, *Proc. IRE* 32, 289 (1944); E. Tonti, *PIER* 32, 1 (2001); K. S. Yee, *IEEE Trans. Antennas Propag.* 14, 302 (1966); U. Vool, M. Devoret, *Int. J. Circuit Theory Appl.* 45, 897 (2017); M. Gell-Mann, *Phys. Rev.* 125, 1067 (1962); D. Marcos et al., *Phys. Rev. Lett.* 111, 110504 (2013); B. D. Josephson, *Phys. Lett.* 1, 251 (1962); K. v. Klitzing et al., *Phys. Rev. Lett.* 45, 494 (1980); J. P. Pekola et al., *Rev. Mod. Phys.* 85, 1421 (2013); BIPM, *SI Brochure*, 9th ed. (2019); R. de-Picciotto et al., *Nature* 389, 162 (1997); L. Saminadayar et al., *Phys. Rev. Lett.* 79, 2526 (1997).

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